

Interim 1 — T2: Code Generation (ER → SQL)

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Slide 1 • Goals

- ER → SQL DDL (tables, PK, FK, indexes)
- Postgres + MySQL dialects
- Configurable ambiguity (inheritance, N:M, weak, multivalued)
- Naming strategy + validation
- LSP + CLI integration
- Tests: unit + golden + negative

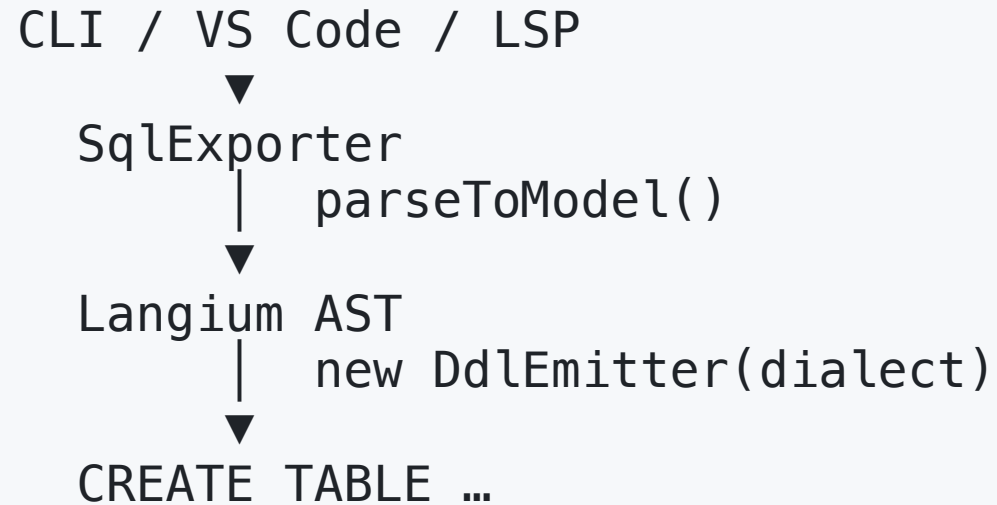
Slide 2 · Research

- bigER's Java `SqlGenerator` + per-dialect `mapDataType`
- Langium AST lifecycle — must use `DocumentBuilder` for `.ref` to resolve
- Textbook ER→relational mapping rules
- Postgres vs MySQL divergence points
- Reference tools (DBML, real SQL parsers) — bigER has bugs

Slide 3 · Status — Done ✓

- `.er` → AST → DDL → file works end-to-end
- Entity → `CREATE TABLE` + PK
- Relationship → bridge table + composite PK + inline FK
- Postgres + MySQL dialect mapping (full category port)
- Derived attrs skipped
- LSP, VS Code command, CLI — all three wired

Slide 4 · Architecture



- `sql-exporter.ts` — orchestrator
- `ddl-emitter.ts` — AST walk
- `dialects.ts` — postgres/mysql + type categories

Slide 5 · Dialect divergence — live example

```
score: DOUBLE    photo: BLOB    bio: CLOB
```

	postgres	mysql
DOUBLE	DOUBLE PRECISION	DOUBLE
BLOB	BYTEA	BLOB
CLOB	TEXT	TEXT

Covered by `dialect-types` fixture, all 3 test stages.

Slide 6 · Testing

- Vitest, **23 green** across Ubuntu / macOS / Windows, Node 22
- Golden files auto-discovered per dialect
- Three stages per fixture:
 - i. **Grammar** — real parsers (`libpg-query` , `dt-sql-parser`)
 - ii. **Exporter** — byte-for-byte diff + parse check
 - iii. **Engine** — Postgres via PGlite (MySQL deferred)

Slide 7 · Demo

1. `examples/test.er` in VS Code → `Export SQL` command
2. CLI: `--dialect postgres` vs `--dialect mysql`
3. `diff` shows only type columns change
4. `yarn test` → 23 green, three stages visible

Slide 8 · Planned — Interim 2

- Explicit `FOREIGN KEY ... ON DELETE/UPDATE`
- Naming strategy (snake_case, prefixes)
- Ambiguity config: inheritance, 1:N, 1:1, weak, multivalued, NULL
- Validation pass (missing types, collisions, risky FKs)
- Config surface: CLI / LSP options / sidecar file
- Docs: mapping rules + feature matrix
- Stretch: Prisma target

Slide 9 · Blockers — feedback please

1. **Cardinality-blind vs -aware** — match bigER or diverge?
2. **Where does config live** — CLI flag / `.er` directive / sidecar?
3. **Dialect abstraction** — scale strategy objects or switch to emitter subclass?
4. **bigER bugs** (e.g. MySQL `CLOB`) — stay faithful or fix? (we fixed for now)
5. **Composition / aggregation → SQL** — FK + warning OK?